

swtpcsim — Changelog

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A simulation of the SWTPC 6800 on the SS-50/SS-30 bus, and the MITS Altair 680b

Changelog

Every release of **swtpcsim**, newest first. Each entry is the short version — what you can do here that you could not do in the release before it. The User Manual describes the program as it is now; this document is the record of how it got there.

0.0.1

The first release — a simulator of **Motorola 6800** machines (the SWTPC 6800 on the SS-50/SS-30 bus, and the MITS Altair 680b), in C++20, with **nothing to fetch**: the TOML parser, the JSON encoder and the line editor are all in-tree, so a fresh clone builds with a C++20 compiler and CMake and no network. SDL3 is optional and detected, never required.

Two machines that boot from a bare command line

- **swtpcsim swtpc** brings up an SWTPC 6800 under **SWTBUG**, its ROM monitor, with an MP-S serial console and a DC-4 floppy controller. **FLEX 2.0** boots from drive 0.
- **swtpcsim altair680** brings up a MITS **Altair 680b** under **MON680**, the 680b's own ROM monitor, on a single 6850 ACIA — and the **KCACR** Kansas City audio-cassette interface loads and saves off tape.
- **swtpcsim cp68** boots **TSC CP/68** from a DC-4 disk.

The boards behind them

A Motorola **6800** CPU; a **memory** board that carries RAM, scratchpad RAM and the monitor ROM as a list of regions; the SWTPC **MP-S** serial console; the Altair 680b **onboard** and **Universal I/O** serial/parallel boards; the **KCACR** cassette; and the SWTPC **DC-4** WD179x floppy controller.

A Motorola **6809** CPU board (**6809**) as well. No built-in machine uses it yet, but a machine file can: **DISASM** and **EDIT** speak 6809, with every indexed mode, and **SHOW BUS IRQ** shows its seven vectors.

The monitor prompt and the debugger

Drive a running machine from the **swtpcsim>** prompt: fit and configure boards (**BOARDS** , **SHOW BOARD** , **SET**), mount disks and tapes, wire serial lines (**CONNECT**), and script a boot with **STARTUP** . The debugger reads the 6800: **DISASM** and the byte-at-a-time **EDIT** assembler speak 6800, **BREAK** / **HISTORY** / watchpoints and conditional breaks (**BREAK ... IF**) stop and trace it, **SYMBOLS** loads a Motorola **as0/as9** listing so disassembly reads by name, and **LOAD** reads a Motorola **S-record** (**.S19**) as well as Intel HEX. **SNAPSHOT** / **RESTORE** save and reload a whole machine.

Driving a machine over MCP

`swtpcsim <machine> --mcp` exposes the machine to an AI assistant as line-delimited JSON-RPC, with typed tools for running, stepping, breakpoints, memory and disassembly rather than a text prompt to screen-scrape. `--mirror` opens a live socket onto the same console, so a person can watch and take over the keyboard. A `run` that will not end can be stopped with a `notifications/cancelled` or a `^C`, and a `status` tool answers at once even while a `run` is in progress. Every tool checks its arguments, so `"from": "0xE0D0"` is refused with the number to send instead of running from 0.

The windows

With SDL3 present, a serial line can open a **built-in terminal window** the simulator draws itself (`CONNECT sio0:a terminal`), and video boards each open their own host window. A headless build refuses the endpoint cleanly rather than opening a line nobody can see.

The package, and the documents in it

Each platform archive holds the program, the **User Manual**, the **monitor** and **debugger** references, this changelog, a cheatsheet, both licences, and `examples/` — `flex`, `cp68`, `altair680` and a `debugger` walkthrough, media included. SDL3 is linked **statically**, so there is nothing to install beside the binary. Unzip it and run it; nothing needs fetching first.